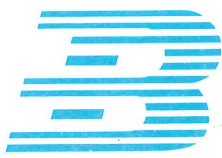


Lecture 1 & 2 with Tutorial



北京工业大学 1 & 2

BEIJING UNIVERSITY OF TECHNOLOGY

100 Ping Le Yuan • Chaoyang District • Beijing 100124 • P.R.China

<http://www.bjut.edu.cn>

1) vector space $\mathbb{R}^n$

$$\min_{x \in \Omega} f(x)$$

12) $x, y \in \mathbb{R}^n$

$$\langle x, y \rangle = \sum_{i=1}^n x_i y_i$$

$$(3) \|x\|_2 = \sqrt{\langle x, x \rangle}$$

induced from $\langle \rangle$

$$(4) \|x\| \cdot \|y\| = |\langle x, y \rangle|$$

$$" = " \Leftrightarrow x \parallel y \quad (x = \alpha y)$$

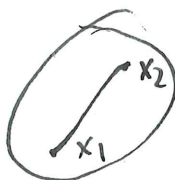
(5) ~~convex set (e.g. C)~~

convex set C

$$\Leftrightarrow \forall x_1^p, x_2^p \in C,$$

$$\forall \alpha \in \boxed{[0,1]} (0,1]$$

$$\alpha x_1 + (1-\alpha)x_2 \in C$$



(6) convex ~~cone~~ cone

$$C \Leftrightarrow \forall \alpha, \beta \geq 0$$

$$\alpha x_1 + \beta x_2 \in C$$

(7) convex ~~hull~~ hull

$\text{conv}(S)$ = the set of ...



(8) half space
hyper plane

$$\{x \mid a^T x = b\}$$



北京工业大学

BEIJING UNIVERSITY OF TECHNOLOGY

100 Ping Le Yuan • Chaoyang District • Beijing 100124 • P.R.China

<http://www.bjut.edu.cn>

$$P \in S_{++}^n \rightarrow P \in R^{n \times n} \quad P^T P, \forall z \in R^n \text{ AND } z \neq 0$$

$$\nexists \quad \Downarrow P > 0 \quad \therefore \exists z^T P z > 0$$

$$\Rightarrow P \in S_{++}^n \Leftrightarrow P = V \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} V^T$$

Example:

$$P = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$$

When $n=2$:

$$P = (V_1, V_2) \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} (V_1^T, V_2^T)^T$$

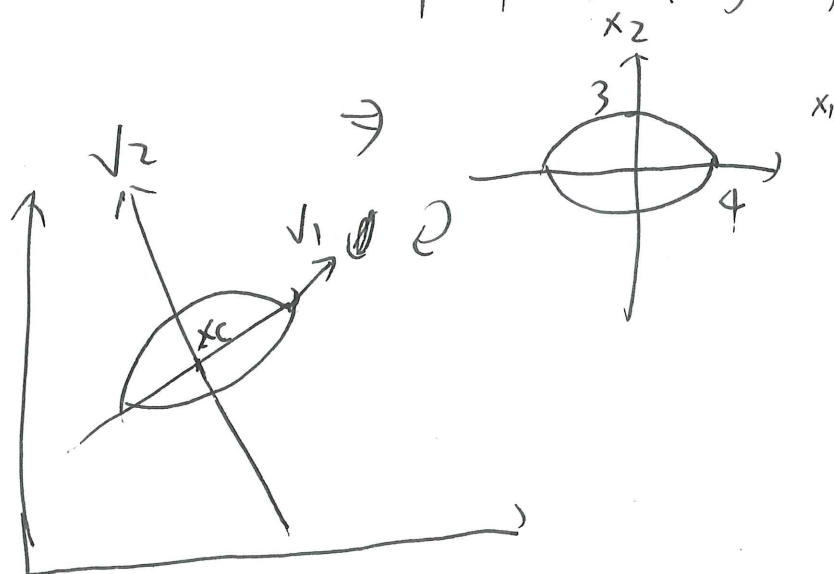
$$P = (V_1, V_2) \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \begin{pmatrix} V_1^T \\ V_2^T \end{pmatrix}$$

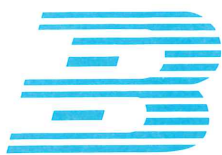
$$\Rightarrow P^{\frac{1}{2}} = (V_1, V_2) \begin{pmatrix} \sqrt{\lambda_1} & 0 \\ 0 & \sqrt{\lambda_2} \end{pmatrix} \begin{pmatrix} V_1^T \\ V_2^T \end{pmatrix}$$

e.g. $P = \begin{pmatrix} 16 & 0 \\ 0 & 9 \end{pmatrix}, \therefore P^{-1} = \begin{pmatrix} \frac{1}{16} & 0 \\ 0 & \frac{1}{9} \end{pmatrix}$

$$x^T C x = 0$$

$$\left\{ x \mid \left(\frac{x_1}{4}\right)^2 + \left(\frac{x_2}{3}\right)^2 \leq 1 \right\}$$





北京工业大学

BEIJING UNIVERSITY OF TECHNOLOGY

100 Ping Le Yuan • Chaoyang District • Beijing 100124 • P.R.China

<http://www.bjut.edu.cn>

$$A = ?$$

$$\Rightarrow A = P^{\frac{1}{2}} = \begin{pmatrix} \sqrt{\lambda_1} & & 0 \\ & \ddots & \\ 0 & & \sqrt{\lambda_n} \end{pmatrix}$$

$$\|x\|_2 = \sqrt{\sum_{i=1}^n x_i^2}$$

$$\|x\|_2 = |x_1| + |x_2| + \dots + |x_n|$$

$$\|x\|_{\infty} = \max_{1 \leq i \leq n} |x_i|$$

$$(A^{\frac{1}{2}})^T A^{\frac{1}{2}}$$

$$X^T A X$$

$$= X^T A^{-\frac{1}{2}} A^{\frac{1}{2}} X$$

$$= (A^{\frac{1}{2}} X)^T A^{\frac{1}{2}} X$$

$$= (A^{\frac{1}{2}} X)^T A^{\frac{1}{2}} X$$

$$= (A^{\frac{1}{2}} X, A^{\frac{1}{2}} X)$$

$$A \in \mathbb{R}^{n \times n} \Leftrightarrow A \preceq A \in S_{++}^n$$

$$\Leftrightarrow A = A^{\frac{1}{2}} A^{\frac{1}{2}}$$

$$\left\{ \begin{pmatrix} x \\ t \end{pmatrix} \mid \sqrt{x_1^2 + x_2^2 + \dots + x_n^2} \leq |t| \right\}$$

$$\|x\|_A = \sqrt{x^T A x}$$

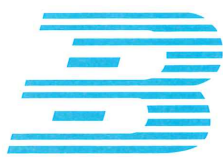
$$A \in \mathbb{R}^{m \times n}$$

$$\Rightarrow A = \begin{bmatrix} \text{columns} \\ \text{rows} \end{bmatrix}$$

$$\mathcal{M} = \{A \in \mathbb{R}^{n \times n} \mid S_A \text{ is convex}\}$$



$$S = \bigcap_{A \in \mathcal{M}} S_A \text{ is convex}$$



北京工业大学

BEIJING UNIVERSITY OF TECHNOLOGY

100 Ping Le Yuan • Chaoyang District • Beijing 100124 • P.R.China

<http://www.bjut.edu.cn>

Tutorial 3

72403895

Chongyu Chang

Question 1

case A: if $x=0$ or $y=0$, it is trivial to be proved.

case B: Assume $\|x\|_p > 0$, and $\|y\|_p > 0$

Now we have: $\|x\|_p = \left[\sum_{i=1}^n |x_i|^p \right]^{\frac{1}{p}}$

Considering " $\frac{1}{p}$ " makes it difficult to analyse, we can use triangle inequality

$\|x+y\|_p \leq \|x\|_p + \|y\|_p$ to prove:

we let $t = \|x\|_p + \|y\|_p$, and $x_1 = \frac{x}{\|x\|_p}$, $x_2 = \frac{y}{\|y\|_p}$

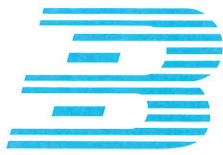
$$\therefore \left\| \frac{\|x\|_p}{t} \cdot x_1 + \frac{\|y\|_p}{t} \cdot y_1 \right\|_p \leq 1$$

$$\therefore \text{let } \lambda = \frac{\|x\|_p}{t}, 1-\lambda$$

~~be a~~
 $\therefore$ Considering $1 = \frac{\|x\|_p}{t} + \frac{\|y\|_p}{t}$,

$$\text{we let } \lambda = \frac{\|x\|_p}{t}, 1-\lambda = \frac{\|y\|_p}{t}$$

$$\therefore \|\lambda x_1 + (1-\lambda) y_1\|_p^p \leq 1$$



北京工业大学

BEIJING UNIVERSITY OF TECHNOLOGY

100 Ping Le Yuan • Chaoyang District • Beijing 100124 • P.R.China

<http://www.bjut.edu.cn>

Now we transfer them to functions, we have:

$$f(\lambda x_1 + (1-\lambda)y_1) \leq \lambda f(x_1) + (1-\lambda)f(y_1)$$

where $f(x) = \|x\|_p^p$ is convex

Considering $f(x)$ is convex,

it is evident that $\|x\|_p = \left[\sum_{i=1}^n |x_i|^p \right]^{\frac{1}{p}}$
is a norm when $p \geq 1$

(I asked some students and schoolmates,
it is it related to Minkowski inequality
or Hölder's inequality?)